

Charotar University of Science and Technology

CHARUSAT Campus, Changa – 388 421.

M Sc (BIOCHEMISTRY)

SEMESTER II EXAMINATION 2011

BC 707: METABOLISM

Date: 5th May 2011

Time: 10.00 AM to 1.00 PM

Total Marks: 70

Instructions to the candidates:

1. Attempt all questions
2. Questions in **Part A** should be answered in the question paper itself. Please do not write your candidate ID number on the question paper.
3. Write answers for Section I and Section II of **Part B** in separate answer sheets.
4. Use of cell phones is strictly prohibited.
5. Use of non-programmable calculators may be allowed if needed.

PART A

Total marks: 20

Q1. Choose the correct option and put ✓ mark in front of it:

- 1) The coenzyme directly concerned with the carboxylation reaction.
a) TPP b) NADP+ c) Biotin d) Pyridoxal Phosphate
- 2) All of the following participate in the mitochondrial pyruvate dehydrogenase reaction except
a) Flavin b) Coenzyme A c) Lipoamide d) Biotin
- 3) All of the following participate in one carbon transfer reaction except _____.
a) Tetrahydrofolate b) Biotin c) s-adenosine methionine d) TPP
- 4) NADP is required as a coenzyme in
a) Glycolysis b) Citric acid cycle c) HMP shunt d) Gluconeogenesis
- 5) Pyruvate Dehydrogenase complex is regulated by :
a) Covalent modification b) Allosteric regulation
c) (a) and (b) both d) Neither of the above
- 6) The first pentose formed in HMP shunt is
(a) Ribose-5-phosphate (b) Ribulose-5-phosphate
(c) Xylose-5-phosphate (d) Xylulose-5-phosphate
- 7) Glucose cannot be synthesized from
(a) Glycerol (b) Alanine (c) Lactate (d) Leucine
- 8) Glycogen synthetase requires as substrate
a) UDP-glucose b) GDP-glucose c) ADP-glucose d) CDP-Glucose
- 9) The formation of malonyl-CoA from acetyl-CoA is catalyzed by:
a) Malonyl-CoA carboxylase b) Acetyl-CoA synthetase
c) Malonyl-CoA synthetase d) Acetyl-CoA carboxylase

- 10) The rate-limiting step in the pathway of cholesterol biosynthesis is catalysed by:
a) HMG-CoA Reductase b) Phosphomevalonate kinase
c) HMG-CoA synthase d) Thiolase
- 11) β -Oxidation of fatty acids requires all the following except
(a) CoA (b) FAD (c) NAD (d) NADP
- 12) Enzyme in fatty acid synthase complex that has pantotheinc acid as coenzyme
a) Acetyl transacylase b) Thioesterase
c) Enoyl Reductase d) Acyl carrier protein
- 13) A yeast protein with methionine at its N terminus typically has a half-life of:
a) More than 20 hours b) Less than 20 hours
c) About 2 minutes d) Less then 2 minutes
- 14) The combined action of an aminotransferase and glutamate dehydrogenase is referred to as :
a) Transamination b) deamination c) Transdeamination
d) All of the above
- 15) These two amino acids are solely ketogenic
a) Isolucine and valine b) Leucine and lysine
c) Glutamate and glutamine d) Proline and Histidine
- 16) Mammals synthesize cysteine from:
a) methionine and serine b) methionine and glycine
c) serine and glycine d) serine and sulfide
- 17) Which of the following bases cannot be directly salvaged?
a) guanine b) hypoxanthine c) xanthine d) adenine
- 18) Which Enzyme causes Reduction of ribonucleotide to deoxy ribonucleotide?
a) ribonucleotide reductase b) nucleoside diphosphate kinase
c) thymidylate synthase d) dUTPase
- 19) In Pyrimidine biosynthesis:
a) bases synthesized piece by piece directly on ribose sugar
b) bases assembled first & then attached to deoxyribose sugar
c) bases assembled first & then attached to ribose sugar
d) bases synthesized piece by piece directly on deoxyribose sugar.
- 20) PRPP synthetase is inhibited by:
a) ATP and GTP b) ADP and GDP c) CTP and UTP d) ATP and CTP

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Part B

Total Marks: 50

Section I

Q1 A Answer Any Two:

- a) Give structure of Carnitine. Explain its role in transport of the fatty acid into mitochondria. [4]
- b) Write short note on Feed back regulation for Cholesterol biosynthesis [4]
- c) How are prostaglandins synthesized [4]

B Ketone bodies are not utilized by Liver. Why? [2]

Q2 A Write Short note on any Two:

- a) Biosynthesis of thymidylate. [3]
- b) Synthesis of Inosine Monophosphate [3]
- c) Pyrimidine biosynthesis [3]

B Discuss about regulation of purine nucleotide biosynthesis [4]

Section II

Q3 A Write short note on:

- a) Mechanism of action of NAD⁺ [5]

OR

- b) Techniques for studying metabolism [5]

B How Biotin involved in carboxylation reaction? Explain with suitable example. [5]

Q4 A Answer Any Two:

- a) List distinctive features of glucokinase and hexokinase [4]
- b) Explain regulation of Citric acid cycle [4]
- c) Explain: How is a synthesis of new glycogen molecule initiated? [4]

B Write following reaction of Glycolysis. [2]

- Dehydration of 2-Phosphoglycerate to Phosphoenolpyruvate
- Oxidation of Glyceraldehyde 3-Phosphate to 1,3-Bisphosphoglycerate

Q5 A What are polyamines? Give Examples. [2]

B Write short notes on: (Any Two)

- a) Transamination reaction [4]
- b) Ubiquitin Tags Proteins for Destruction [4]
- c) Biosynthesis of Tryptophan from chorismate [4]